# Local Information, Global Structure: Pigeonhole, Ramsey, Radon, and Helly

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## §1.1 Pigeonhole as Averaging

If $N$ objects are placed into $r$ boxes, some box contains at least

$$\left\lceil \frac{N}{r} \right\rceil$$

objects. The proof is the contrapositive: if every box held at most $q$ objects, the total would be at most $rq$.

The theorem is easy; choosing the boxes is the real work. Suppose nine elements are each recorded by membership in three sets $A, B, C$. Every element receives a bit pattern

$$(\mathbf{1}_A(x), \mathbf{1}_B(x), \mathbf{1}_C(x)) \in \{0,1\}^3.$$

There are only

$$2^3 = 8$$

possible patterns. Among nine elements, two have the same pattern and therefore belong to exactly the same subcollection of $A, B, C$.

The conclusion is not merely "two objects coincide." A carefully chosen finite state space forces two complicated objects to become indistinguishable with respect to all recorded tests.

## §1.2 Partial Sums and Divisibility

Let $a_1, \dots, a_n$ be integers. There is always a nonempty consecutive block whose sum is divisible by $n$.

Consider the partial sums

$$s_k = a_1 + \cdots + a_k, \qquad k = 1, \dots, n,$$

and reduce them modulo $n$. If some

$$s_k \equiv 0 \pmod{n},$$

then $a_1 + \cdots + a_k$ is the required block. Otherwise the $n$ partial sums occupy only the $n-1$ nonzero residue classes, so two satisfy

$$s_i \equiv s_j \pmod{n}, \qquad i < j.$$

Subtracting gives

$$a_{i+1} + \cdots + a_j = s_j - s_i \equiv 0 \pmod{n}.$$

The objects are partial histories, the boxes are residue states, and a collision cancels the common prefix. This pattern appears throughout additive combinatorics and automata: encode a running process in finitely many states, then use repetition to extract a structured segment.

## §1.3 Pigeonhole and Sequential Compactness

Let $(x_n)$ be an infinite sequence in $[0,1]$. Divide the interval into two closed halves. At least one half contains infinitely many terms; call it $I_1$. Divide $I_1$ into two halves and choose one, $I_2$, that again contains infinitely many terms. Continuing gives nested intervals

$$I_1 \supset I_2 \supset I_3 \supset \cdots$$

with

$$|I_k| = 2^{-k}.$$

Choose indices

$$n_1 < n_2 < \cdots$$

so that

$$x_{n_k} \in I_k.$$

The nested interval theorem gives a unique point

$$x \in \bigcap_{k=1}^{\infty} I_k.$$

For $j \geq k$, both $x_{n_j}$ and $x$ lie in $I_k$, so

$$|x_{n_j} - x| \leq 2^{-k}.$$

Hence

$$x_{n_j} \to x.$$

This is Bolzano–Weierstrass on $[0,1]$. The infinite conclusion is built by iterating a finite pigeonhole step at finer and finer scales. In a compact metric space, finite covers by small balls play the role of interval subdivisions.

## §1.4 Ramsey's Forced Patterns

Color every edge of the complete graph $K_6$ red or blue. Choose a vertex $v$. Among the five edges incident to $v$, at least three have the same color; suppose

$$va, vb, vc$$

are red.

If any of

$$ab, bc, ca$$

is red, that edge together with $v$ forms a red triangle. If none is red, then all three are blue and $abc$ is a blue triangle. Thus every red–blue coloring of $K_6$ contains a monochromatic triangle:

$$R(3,3) \leq 6.$$

The bound is sharp. Color the edges of a five-cycle red and its five diagonals blue. The red graph is a cycle and has no triangle; the blue graph is another five-cycle and also has no triangle. Therefore

$$R(3,3) > 5,$$

so

$$\boxed{R(3,3) = 6.}$$

Pigeonhole forced three edges of one color around a vertex. The graph structure then amplified that local repetition into a monochromatic triangle. Ramsey theory studies how large an ambient configuration must be before a specified organized substructure becomes unavoidable.

## §1.5 Intervals give the one-dimensional Helly theorem

Let

$$I_i = [a_i, b_i] \qquad (i = 1, \dots, m)$$

be closed intervals, and assume every pair intersects. Set

$$a = \max_i a_i, \qquad b = \min_i b_i.$$

Choose indices $p, q$ with

$$a = a_p, \qquad b = b_q.$$

Since $I_p \cap I_q \neq \varnothing$,

$$a_p \leq b_q,$$

so $a \leq b$. Every interval contains the segment $[a, b]$, and therefore

$$\bigcap_{i=1}^{m} I_i \neq \varnothing.$$

Pairwise intersection is enough in one dimension because the order lets one reduce all left endpoints to their maximum and all right endpoints to their minimum. In higher dimensions there is no total order of boundaries; convexity and affine dependence replace it.

## §1.6 Radon's Theorem from Affine Dependence

**Theorem 1.6.1** (Radon)

Any $d+2$ points in $\mathbb{R}^d$ can be partitioned into two disjoint sets whose convex hulls intersect.

Let the points be $p_1, \dots, p_{d+2}$. The lifted vectors

$$(p_i, 1) \in \mathbb{R}^{d+1}$$

are linearly dependent, so there are scalars $\lambda_i$, not all zero, with

$$\sum_i \lambda_i p_i = 0, \qquad \sum_i \lambda_i = 0.$$

Let

$$I = \{i : \lambda_i > 0\}, \qquad J = \{i : \lambda_i < 0\}.$$

Both are nonempty. Put

$$s = \sum_{i \in I} \lambda_i = -\sum_{j \in J} \lambda_j > 0.$$

Then

$$\sum_{i \in I} \frac{\lambda_i}{s} p_i = \sum_{j \in J} \frac{-\lambda_j}{s} p_j.$$

Both sides are convex combinations. Their common value lies in

$$\operatorname{conv}\{p_i : i \in I\} \cap \operatorname{conv}\{p_j : j \in J\}.$$

The Radon partition is the geometric form of a minimal affine dependence. In oriented-matroid language, such a signed dependence is the prototype of a circuit.

## §1.7 Helly's Theorem via Radon

**Theorem 1.7.1** (Finite Helly)

Let $C_1, \dots, C_m$ be convex subsets of $\mathbb{R}^d$, with $m \geq d+1$. If every $d+1$ of them have nonempty intersection, then

$$\bigcap_{i=1}^{m} C_i \neq \varnothing.$$

We prove the theorem by induction on $m$. The case $m = d+1$ is the hypothesis. Assume $m > d+1$ and the result holds for $m-1$ sets.

For each $i = 1, \dots, d+2$, the family obtained by omitting $C_i$ has $m-1$ members and still satisfies the local intersection hypothesis. By induction, choose

$$x_i \in \bigcap_{j \neq i} C_j.$$

Apply Radon's theorem to $x_1, \dots, x_{d+2}$. There is a partition

$$I \sqcup J = [d+2]$$

and a point

$$x \in \operatorname{conv}\{x_i : i \in I\} \cap \operatorname{conv}\{x_j : j \in J\}.$$

Fix a set $C_k$. If $k \in I$, then every $x_j$ with $j \in J$ belongs to $C_k$, because $j \neq k$. Convexity gives

$$x \in C_k.$$

If $k \in J$, use the convex hull indexed by $I$. If $k > d+2$, every $x_i$ lies in $C_k$. Thus $x$ belongs to every $C_k$, completing the induction.

The proof shows exactly where the number $d+1$ comes from: Radon's affine dependence begins at $d+2$ points.

## §1.8 Carathéodory's Reduction

**Theorem 1.8.1** (Carathéodory)

If $x \in \operatorname{conv} S \subset \mathbb{R}^d$, then $x$ lies in the convex hull of at most $d+1$ points of $S$.

Write

$$x = \sum_{i=1}^{m} \alpha_i p_i, \qquad \alpha_i > 0, \qquad \sum_i \alpha_i = 1,$$

with $m > d+1$. The points are affinely dependent, so there are $\lambda_i$, not all zero, satisfying

$$\sum_i \lambda_i p_i = 0, \qquad \sum_i \lambda_i = 0.$$

After changing all signs if necessary, some $\lambda_i > 0$. Let

$$t = \min_{\lambda_i > 0} \frac{\alpha_i}{\lambda_i}.$$

Then the new coefficients

$$\alpha_i' = \alpha_i - t\lambda_i$$

remain nonnegative, still sum to one, and give the same point $x$. At least one coefficient becomes zero. Repeating removes points until at most $d+1$ remain.

Helly compresses an intersection certificate to small subfamilies. The preceding convex-hull theorem compresses a representation to at most $d+1$ points. Both reductions are driven by affine dependence.

## §1.9 Fractional Helly tolerates failed local tests

The ordinary Helly theorem assumes that every $(d+1)$-tuple intersects. Fractional Helly allows failures.

For each $d$ and $\alpha > 0$, there is $\beta = \beta(d, \alpha) > 0$ such that the following holds. If at least an $\alpha$-fraction of the $(d+1)$-tuples in a family of $n$ convex sets in $\mathbb{R}^d$ have nonempty intersection, then some point belongs to at least $\beta n$ members of the family. One may take

$$\beta = 1 - (1-\alpha)^{1/(d+1)}.$$

In one dimension, the sets are intervals. Intersecting pairs are edges of an interval graph, while intervals containing a common point form a clique. If a positive fraction of all pairs intersect, the theorem guarantees a clique containing a positive fraction of all intervals. Thus numerous local successes cannot be dispersed arbitrarily; they concentrate around a common witness.

The conclusion is weaker than total intersection but robust under noise. It is an extremal version of Helly rather than a different slogan about local-to-global behavior.

## §1.10 Three disks and one missing triangle

For a finite family of sets $\mathcal{F} = \{U_i\}_{i \in I}$, its *nerve* is the simplicial complex

$$N(\mathcal{F}) = \left\{ J \subseteq I : \bigcap_{j \in J} U_j \neq \varnothing \right\}.$$

Vertices record sets, edges record pairwise intersections, and a filled triangle appears precisely when three sets share a point. The nerve theorem says that if the sets form a suitable open *good cover*—every nonempty finite intersection is contractible—then

$$\boxed{\bigcup_{i \in I} U_i \simeq N(\mathcal{F})}$$

up to homotopy equivalence. For convex sets the intersection condition is automatic. This is the topological mechanism behind Helly-type theorems: information about intersections of small subfamilies constrains which simplices, and hence which holes, the nerve may have.

Here is also a short homological explanation of the number $d+1$ in a topological version of Helly's theorem. Suppose a finite good cover by open subsets of $\mathbb{R}^d$ has at least $d+1$ members, every $d+1$ members meet, yet the whole family has empty intersection. Choose a *smallest* subfamily with empty intersection, of size $m \geq d+2$. Every proper subfamily meets, so its nerve is the boundary of an $(m-1)$-simplex, a sphere $S^{m-2}$. The nerve theorem would then give

$$\widetilde{H}_{m-2}\left(\bigcup_{i=1}^{m} U_i\right) \cong \widetilde{H}_{m-2}(S^{m-2}) \neq 0.$$

But an open subset of $\mathbb{R}^d$ has no nonzero homology in degrees $j \geq d$. Therefore $m-2 < d$, or $m \leq d+1$. In particular, if every $d+1$ members meet, no empty intersection can exist. Convex families are good covers, so classical Helly is included.

A small geometric experiment makes the nerve theorem almost visible. Put three centers at the vertices of an equilateral triangle of side 1,

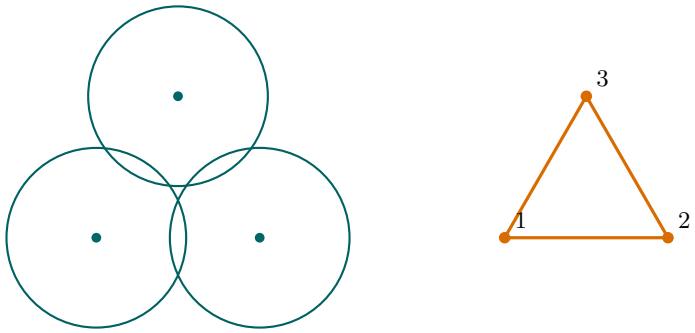
$$p_1 = (0,0), \qquad p_2 = (1,0), \qquad p_3 = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right),$$

and draw open disks $U_i = B(p_i, r)$ of the same radius.

Two disks meet as soon as $2r > 1$, so $r > \frac{1}{2}$ ensures *every pair* intersects. But a point can belong to all three disks only if its maximum distance from the three centers is less than $r$. Symmetry, or a short smallest-enclosing-circle argument, shows that the least possible maximum distance is the circumradius $1/\sqrt{3}$. Thus if

$$\boxed{\frac{1}{2} < r < \frac{1}{\sqrt{3}},}$$

every pair intersects but the triple intersection is empty. Take $r = 11/20$.




three overlapping disks their nerve: an unfilled triangle

Figure 1.1: All three pairwise intersections are nonempty, so every nerve edge is present. The missing triple intersection leaves out the two-dimensional face.

The nerve is exactly the boundary of a triangle:

$$N(\{U_1, U_2, U_3\}) = \{\varnothing, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}\}.$$

Since intersections of disks in the plane are convex whenever nonempty, this is a good cover. The nerve theorem now says that the union $U_1 \cup U_2 \cup U_3$ has the homotopy type of a circle. Its central hole is recorded by the one-dimensional homology of the *missing face*.

At $r > 1/\sqrt{3}$, the triple intersection becomes nonempty. The nerve gains the triangle $\{1, 2, 3\}$, and that cycle becomes the boundary of a 2-simplex. An intersection condition that changes at one explicit radius has removed a topological hole. For open disks the endpoint $r = 1/\sqrt{3}$ still lacks the triple intersection; the distinction between open and closed balls matters exactly at this threshold.